Does Hardware Change Which Generative Model Is Efficient?

Evaluating Diffusion and Flow-Matching Samplers Across Heterogeneous GPUs

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Advanced Topics in Artificial Intelligence, 2026–2027 | October 2026

Abstract

This project asks whether efficiency rankings of image generators survive a change of hardware. I will compare pretrained diffusion and flow-matching models with a shared transformer backbone, using sampling settings that can be evaluated within one semester. For each available GPU, I will construct quality–cost Pareto frontiers from Fréchet Inception Distance (FID), precision/recall, latency, memory, and energy per image. The main outcome will be a reproducible test of when the number of function evaluations (NFE) is a reliable efficiency proxy and when hardware changes the preferred configuration.

1. Research question and hypotheses

The central question is: *Do diffusion and flow-matching sampler rankings remain stable across GPU classes and numerical precisions?* Three sub-questions guide the study:

- How well does NFE predict latency and energy on each device?
- Do quality–latency Pareto frontiers preserve their ordering across devices?
- At equal measured cost, which model family loses quality more slowly as the step count or precision is reduced?

I expect NFE to correlate with latency within one GPU and sampler, but to miss solver overhead, memory-bandwidth effects, and hardware-specific precision support. Consequently, some near-frontier configurations may exchange rank between devices. I also expect the flow-matching model to degrade more gracefully in the low-step regime because its learned paths are designed to be straighter. These are testable hypotheses; stable rankings would instead support NFE as an adequate proxy within the tested range.

2. Motivation and related work

Diffusion models generate strong images through iterative denoising [2], while flow matching learns a continuous velocity field along simpler probability paths [4]. DiT [8] and SiT [6] are a useful controlled pair: they use closely matched transformer backbones but different training objectives. Fast diffusion solvers such as DDIM and DPM-Solver reduce sampling to tens of evaluations [9, 5]; however, papers commonly report NFE or timings on one accelerator. FLOPs and parameter counts are known to disagree with measured efficiency [1], and evaluation details can materially change FID [7]. The missing evidence is whether diffusion-versus-flow conclusions transfer across hardware under the same measurement protocol.

3. Methodology

Models and factors. I will use public ImageNet-256 checkpoints for one DiT and one comparable SiT configuration. The core sweep uses two representative samplers per family (DDIM and DPM-Solver for DiT; Euler and Heun ODE for SiT), three step counts (8, 16, 32), two precisions (bf16 and fp32), and batch sizes 1 and 8. This gives 24 model–sampler–step–precision configurations, with batch size varied only for timing. No model is trained from scratch. If checkpoint compatibility requires adaptation, fine-tuning will be limited to a small parameter-efficient pilot rather than a full training run.

Hardware and protocol. Measurements will use three device classes: the available local GPU, a free-tier NVIDIA T4 on Google Colab, and the Ikaros nodes equipped with two NVIDIA GH200 GPUs; each run uses one GPU so that results remain comparable. The Colab T4 is used for timing sweeps only and adds an older architecture without native bf16 support. I will pin software versions, use fixed seeds and class labels, warm up kernels, and report medians and interquartile ranges from repeated CUDA-event timings. Peak memory and energy will be collected through NVML. Compilation settings and classifier-free guidance will be fixed, and all network evaluations will count toward NFE.

Quality evaluation. Quality is measured once per model configuration using the same generated samples and pipeline, because repeating identical large image sets on every GPU would waste limited wall time. I will use clean-fid, fixed ImageNet reference statistics, and improved precision/recall [3]. A pilot uses 2,000 images per configuration; only configurations on or near the preliminary frontier are promoted to 10,000 images and three seeds. Selected outputs will be reproduced on each device to check that numerical differences do not materially change quality.

4. Analysis, feasibility, and expected contribution

For each device, I will regress latency and energy on NFE and inspect sampler-specific residuals. I will build FID–latency and FID–energy Pareto frontiers, compare full rankings with Kendall's τ, and flag any configuration

that is efficient on one device but dominated on another. For the low-step question, I will compare the increase in FID at matched measured cost rather than at matched NFE.

The staged design keeps the project feasible under restricted Ikaros wall time: expensive image generation is concentrated on one GPU, while cross-device work consists mainly of short timing sweeps and validation runs. If access is tighter than expected, I will retain both model families and all three step counts, but drop fp32 quality generation and evaluate only batch size 1 on Ikaros. The deliverables are (i) a reproducible benchmark script and configuration manifest, (ii) hardware-specific Pareto plots and rank-agreement statistics, and (iii) evidence-based guidance on when NFE is a trustworthy proxy. The scope is deliberately limited to inference-time evaluation of pretrained models; serving-system design, training from scratch, and exhaustive solver searches are outside the project.

References

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